Chenxu Liu 刘宸旭

B.Sc. in Physics — Shandong University, Qingdao

202422201069@mail.sdu.edu.cn · physicsliu299792458@outlook.com · github.com/PhyLiusir
Qingdao, China
Chinese (native) · English (working proficiency)

EDUCATION

Shandong University, Qingdao
2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Physics as the toolbox — statistical mechanics, quantum field theory, numerical methods — with the working hours on large models and agents instead: read, reproduce at small scale, then change one thing and watch what breaks.

Self-directed — large models and agents
2024 – now

Serving and post-training pipelines reproduced at small scale; agent tooling rewritten against real task batches; each result kept as a log with its script.

HOW I WORK

Derive from first principles, not from precedent — ugly is fine, fabricated is not. Every claim has to replay from a log.

RESEARCH INTERESTS

Large models: deployment and serving economics, post-training (SFT → preference → RL), and new attention architectures — Loop Attention.
Agents: agent construction, tool design, multi-agent collaboration, retrieval-augmented generation.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Deployment (LM 01).** Cost structure of one model across serving stacks, batching and precision — cost per token, P99 latency and peak memory read together.
- **Post-training (LM 02).** A full SFT → preference → RL chain reproduced at small scale, then ablated stage by stage; focus on which gains survive on a smaller model.
- **Loop Attention (LM 03).** Reusing one attention block across depth — memory, wall-clock and training stability measured against a stacked baseline. Main line.
- **Agent construction (AG 01).** The smallest useful combination of loop, memory and planning; a taxonomy of tool-less agent failures as the starting object.
- **Tool design (AG 02).** Schemas, error surfaces and return shapes for a batch of real tasks, comparing failure distributions before and after.
- **Multi-agent collaboration (AG 03).** A strong single-agent baseline first; multi-agent setups have to beat it reliably before roles get split.
- **RAG (AG 04).** A citation-required QA set and a scoring script: an answer must land on a specific passage or it counts as wrong.

CURRENT WORK

- **LM 03 · Loop Attention — the bill for depth reuse.** memory, wall-clock and training stability measured against a stacked baseline.
- **LM 01 · Serving comparison log.** throughput, memory and tail latency for one model across serving and quantisation configs.
- **AG 02 · Tool interface.** schemas and error handling redone for a batch of real tasks, with the failure modes logged.
- **AG 04 · Citation-required RAG eval set.** an answer has to land on a specific passage, or it counts as wrong.

OPEN QUESTIONS

- **LM 01 · Deployment.** At what load does the batching-versus-tail-latency trade-off flip sign? · Is the throughput bought with quantisation worth the accuracy it costs? · Should the KV-cache memory budget be accounted per request or per tenant?
- **LM 02 · Post-training.** Which data-selection metric actually predicts downstream gain? · What observable signals reveal that an eval set has been contaminated? · At what model size do RL-stage gains start to appear?
- **LM 03 · New attention architecture — Loop Attention.** Where is the trade-off point between depth reuse and width scaling? · Which metric detects the representation collapse caused by reuse, early? · On long-context tasks, could reuse turn from cost into advantage?
- **AG 01 · Building agents.** Which failure modes can extra tools fix, and which can they not? · Who should decide when a memory gets written? · At what task length does planning start to pay for itself?
- **AG 02 · Tool design.** How detailed must an error message be before the model actually changes behaviour? · Is it better for tools to be coarse-grained or fine-grained? · How structured should a return value be?
- **AG 03 · Multi-agent collaboration.** How does coordination overhead grow with the number of roles? · What signal says roles should be merged rather than split further? · How does failure propagate between roles?
- **AG 04 · RAG.** In what range is the chunk-size versus recall relationship most stable? · Should a wrong citation cost more than no citation at all? · At what corpus size does re-ranking start to pay off?

TECHNICAL SKILLS

LANGUAGES	Python (primary), C++ and CUDA, Bash, Git and code review
MODEL SIDE	PyTorch training and fine-tuning, transformers and custom layers, serving and quantisation, LoRA and preference alignment
AGENT SIDE	Tools and function calling, multi-agent orchestration, vector retrieval and RAG, evaluation sets and scoring scripts
TOOLING	Linux and containers, distributed training basics, reproducible experiments, technical writing